**Example 12.9**

Verify (i) closure property, (ii) commutative property, (iii) associative property, (iv) existence of identity, and (v) existence of inverse for the operation $+_5$ on $\mathbb{Z}_5$ using table corresponding to addition modulo 5.

**Solution**

It is known that $\mathbb{Z}_5 = \{[0],\ [1],\ [2],\ [3],\ [4]\}$. The table corresponding to addition modulo 5 is as follows: We take reminders $\{0,1,2,3,4\}$ to represent the classes $\{[0],[1],[2],[3],[4]\}$.

| $+_5$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 2 | 3 | 4 |
| 1 | 1 | 2 | 3 | 4 | 0 |
| 2 | 2 | 3 | 4 | 0 | 1 |
| 3 | 3 | 4 | 0 | 1 | 2 |
| 4 | 4 | 0 | 1 | 2 | 3 |

Table12.2

(i) Since each box in the table is filled by **exactly one element** of $\mathbb{Z}_5$, the output $a +_5 b$ is unique and hence $+_5$ is a **binary operation**.

(ii) The entries are **symmetrically** placed with respect to the **main diagonal**. So $+_5$ has **commutative property**.

(iii) The table cannot be used directly for the verification of the associative property. So it is to be verified as usual.

For instance, $(2 +_5 3) +_5 4 = 0 +_5 4 = 4 \pmod 5$

and $2 +_5 (3 +_5 4) = 2 +_5 2 = 4 (\mathrm{mod}\, 5)$.

Hence $(2 +_5 3) +_5 4 = 2 +_5 (3 +_5 4)$.

Proceeding like this one can verify this for all possible triples and ultimately it can be shown that $+_5$ is associative.

(iv) The row headed by 0 and the column headed by 0 are identical. Hence the identity element is 0.

(v) The existence of inverse is guaranteed provided the identity 0 exists in each row and each column. From Table12.2, it is clear that this property is true in this case. The method of finding the inverse of any one of the elements of $\mathbb{Z}_5$, say 2 is outlined below.

First find the position of the identity element 0 in the III row headed by 2. Move horizontally along the III row and after reaching 0, move vertically above 0 in the IV column, because 0 is in the III row and IV column. The element reached at the topmost position of IV column is 3. This element 3 is nothing but the inverse of 2, because, $2 +_5 3 = 0 (\mathrm{mod}\, 5)$. In this way, the inverse of each and every element of $\mathbb{Z}_5$ can be obtained. Note that the inverse of 0 is 0,that of 1 is 4, that of 2 is 3, that of 3 is 2 , and, that of 4 is 1.

**Example 12.10**

Verify (i) closure property, (ii) commutative property, (iii) associative property, (iv) existence of identity, and (v) existence of inverse for the operation $\times_{11}$ on a subset $A = \{1,3,4,5,9\}$ of the set of remainders $\{0,1,2,3,4,5,6,7,8,9,10\}$.



12th_Maths_Vol 2_EM_CH 12_Discrete Mathematics.indd 234 16-12-2022 16:54:25